B.Sc. Engg. CSE 6th Semester | 15 February 2023 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION | **SUMMER SEMESTER, 2021-2022**

DURATION: 1 HOUR 30 MINUTES | **FULL MARKS: 50**

CSE 4615: Wireless Networks

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Define *Access Networks* along with a general taxonomy. 2.5

b) Conduct a comparison between wired and wireless networks based on the following parameters. 3.5

- Standard
- Medium access principle
- Interference and fluctuations vulnerability
- Signal attenuation
- Throughput efficiency
- Delay
- Reliability
- Flexibility
- Maintenance cost and time

c) Answer the following questions: 2.5 + 3.5 (CO1) (PO2)

i. Can Frequency Division Multiple Access (*FDMA*) and Time Division Multiple Access (*TDMA*) be used together? Explain the concept with appropriate diagram.

ii. Mention the effectiveness of combined Code Division Multiple Access (*CDMA*), and Space Division Multiple Access (*SDMA*) scheme in high density wireless networks.

d) Identify the purpose of following interframe spaces (*IFSs*) in data transmission paradigm. 2.5 × 2

i. Short Inter-frame Space (SIFS)

ii. PCF Inter-frame Space (PIFS)

2. a) Mention the effectiveness of using duplex radio for collision detection in wireless networks. 3 (CO2) (PO2)

b) Consider a Basic Service Set (*BSS*) of *IEEE 802.11* wireless LAN (*WLAN*) consisting of three stations *A*, *B*, and *C*. Assume that all the stations compete for accessing a shared channel by means of Distributed Coordination Function (*DCF*) and they can all hear each other's transmission. 8 (CO3) (PO1)

Draw a Timeline Diagram representing the sequence of actions for one successful re-transmission of three *MPDUs* (fragmented portion of a single MAC Service Data Unit) from Station B to Station C.

The diagram should depict the detailed backoff procedure conducted by all the stations in this *BSS*. Consider the minimum Contention Window, (CW_{min}) value is 7. Consider the *RTS/CTS* transmission is enabled in this scenario.

Note that, the *x*-axis of the diagram shows time and *y*-axis shows one horizontal line for each of the contending stations. An action (i.e., transmission of a frame) is represented by a horizontal bold line where the line is placed in the same horizontal line of the station with line length representing period.

c) Random Access based multiple-access protocols in wireless networks result in poor channel utilization. Along with proper argument and diagram, identify the root cause(s) behind such inefficiency. 5 (CO1) (PO2)

3. a) What is the difference between *Multiplexing* and *Multiple Access* technique? 3

b) How does a newly arrived station join an existing *Independent Basic Service Set* (*IBSS*) of *IEEE 802.11 WLAN*? Mention the role of *Beacon* and *Probe Response frames* in this process. 5 (CO3) (PO1)

c) Consider a Quality of Service (*QoS*) supporting Basic Service Set (*QBSS*) of *IEEE 802.11e* wireless LAN (*WLAN*) consists of three stations (*A*, *B*, and *C*). Assume that all the stations compete for accessing a shared channel by means of *EDCA*, a contention-based Hybrid Coordination Function (*HCF*) required for enabling time prioritized *QoS* services. 9 (CO3) (PO1)

Each of the stations of this QBSS maintains two Access Categories (*AC*) labeled as *AC_VO* (voice), and *AC_BK* (background). Note here that, the *AC_VO* and *AC_BK* serves the high priority voice data and low priority background data of the applications running on *QBSS*.

Minimum Contention Window (CW_{min}) values for two Access Categories are listed in Table 1. Arbitration Interframe Space *AIFS[AC]* for two *AC* are listed in Table 2.

The size of the *CW* in backoff stage *i*, after *i-1* times unsuccessful transmission is determined by the following formula.

$$CWi[AC] = min[2^i(CWmin[AC] + 1)\text{-}1, CWmax[AC]]$$

Table 1: Contention Window values for Question 3.c)

Item	*Value*
CW_{min} *[AC_VO]*	4
CW_{min} *[AC_BK]*	6
CW_{max} *[AC_VO]*	16
CW_{max} *[AC_BK]*	36

Table 2: Arbitration Interframe Space values for Question 3.c)

Item	*Value*
AIFS[AC_VO]	*SIFS* + 2 * a *Slot Time*
AIFS[AC_BK]	*SIFS* + 4 * a *Slot Time*

Now to clarify the concept of QoS assurance by *IEEE 802.11e EDCA*, draw a Timeline Diagram representing the sequence of actions for successful transmissions of one voice data and one background data respectively from station *A* and *B* both destined to station *C*.

The diagram should depict the detailed backoff procedure performed by all the stations in the *QBSS*. Consider the *RTS/CTS* transmission is disabled in this scenario.

Note that the x-axis of the diagram shows time and y-axis shows one horizontal line for each of the backoff entities of contending stations. An action (i.e., transmission of a frame) is represented by a horizontal bold line where the line is placed in the same horizontal line of the station with line length representing period.